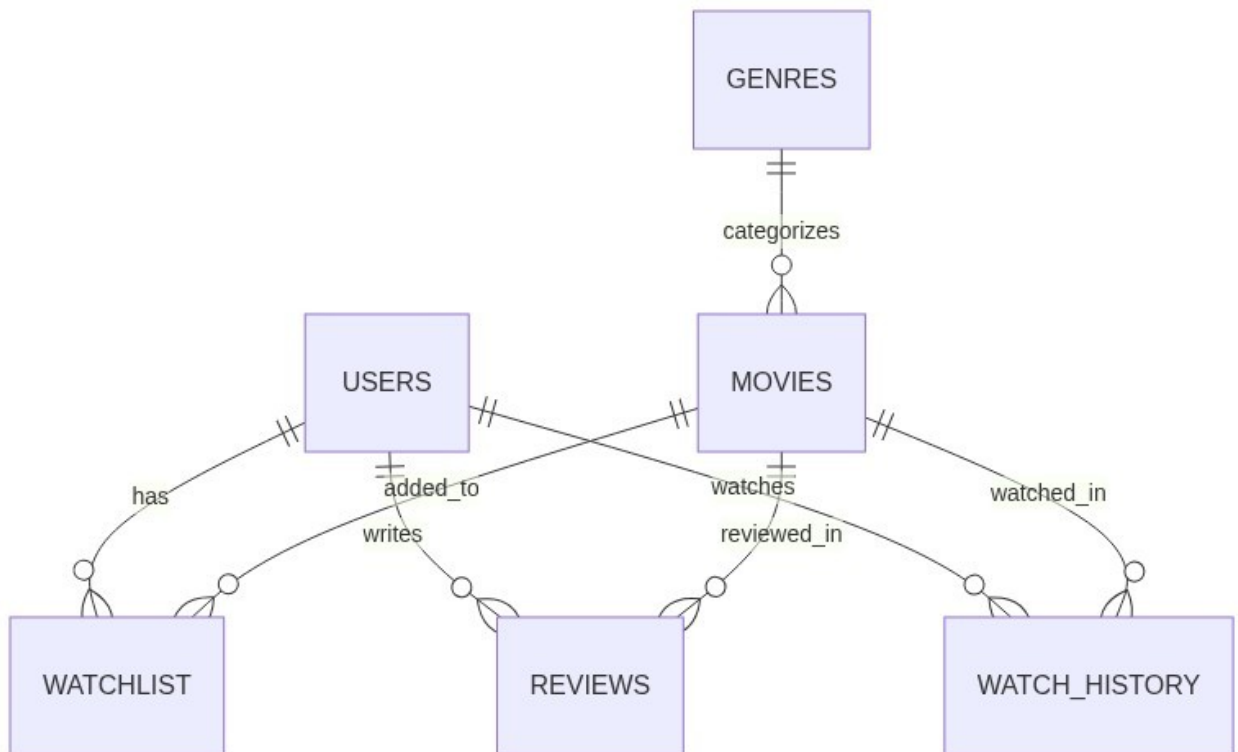


MovieFlix / Movie Recommendation App - ER Diagrams

Database Design Documentation

1. Implementation ER Diagram (Actual Project Database)

This ER diagram represents the actual database schema implemented in the Movie Recommendation App project (movie_recommender database).



1.1 Entity Definitions

USERS

Column	Type	Key	Description
id	INT	PK	Auto-increment primary key
fullname	VARCHAR(100)		User full name

email	VARCHAR(100)	UK	Unique email address
password	VARCHAR(255)		Hashed password (bcrypt)
created_at	TIMESTAMP		Account creation timestamp

GENRES

Column	Type	Key	Description
id	INT	PK	Auto-increment primary key
genre_name	VARCHAR(50)	UK	Genre name (Action, Comedy, etc.)

MOVIES

Column	Type	Key	Description
id	INT	PK	Auto-increment primary key
title	VARCHAR(255)		Movie title
description	TEXT		Movie synopsis/description
genre_id	INT	FK	References GENRES.id
release_year	YEAR		Year of release
duration	INT		Duration in minutes
rating	DECIMAL(3,1)		Average rating (e.g., 8.5)
poster_url	VARCHAR(500)		Poster image URL
trailer_url	VARCHAR(500)		Trailer video URL
created_at	TIMESTAMP		Record creation timestamp

WATCHLIST

Column	Type	Key	Description
id	INT	PK	Auto-increment primary key
user_id	INT	FK	References USERS.id (ON DELETE CASCADE)
movie_id	VARCHAR(30)		TMDB movie ID (external reference)
movie_title	VARCHAR(255)		Cached movie title
poster	VARCHAR(500)		Cached poster path

created_at	TIMESTAMP		When added to watchlist
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REVIEWS

Column	Type	Key	Description
user_id	INT	FK, PK	References USERS.id (composite PK)
movie_id	VARCHAR(30)	FK, PK	TMDB movie ID (composite PK)
movie_title	VARCHAR(255)		Cached movie title
rating	DECIMAL(2,1)		User rating (1-10)
review	TEXT		Review text content
created_at	TIMESTAMP		Review creation timestamp

WATCH_HISTORY

Column	Type	Key	Description
user_id	INT	FK, PK	References USERS.id (composite PK)
movie_id	VARCHAR(30)	FK, PK	TMDB movie ID (composite PK)
title	VARCHAR(255)		Cached movie title
poster	VARCHAR(500)		Cached poster path
watched_at	TIMESTAMP		Last watched timestamp (auto-updated)

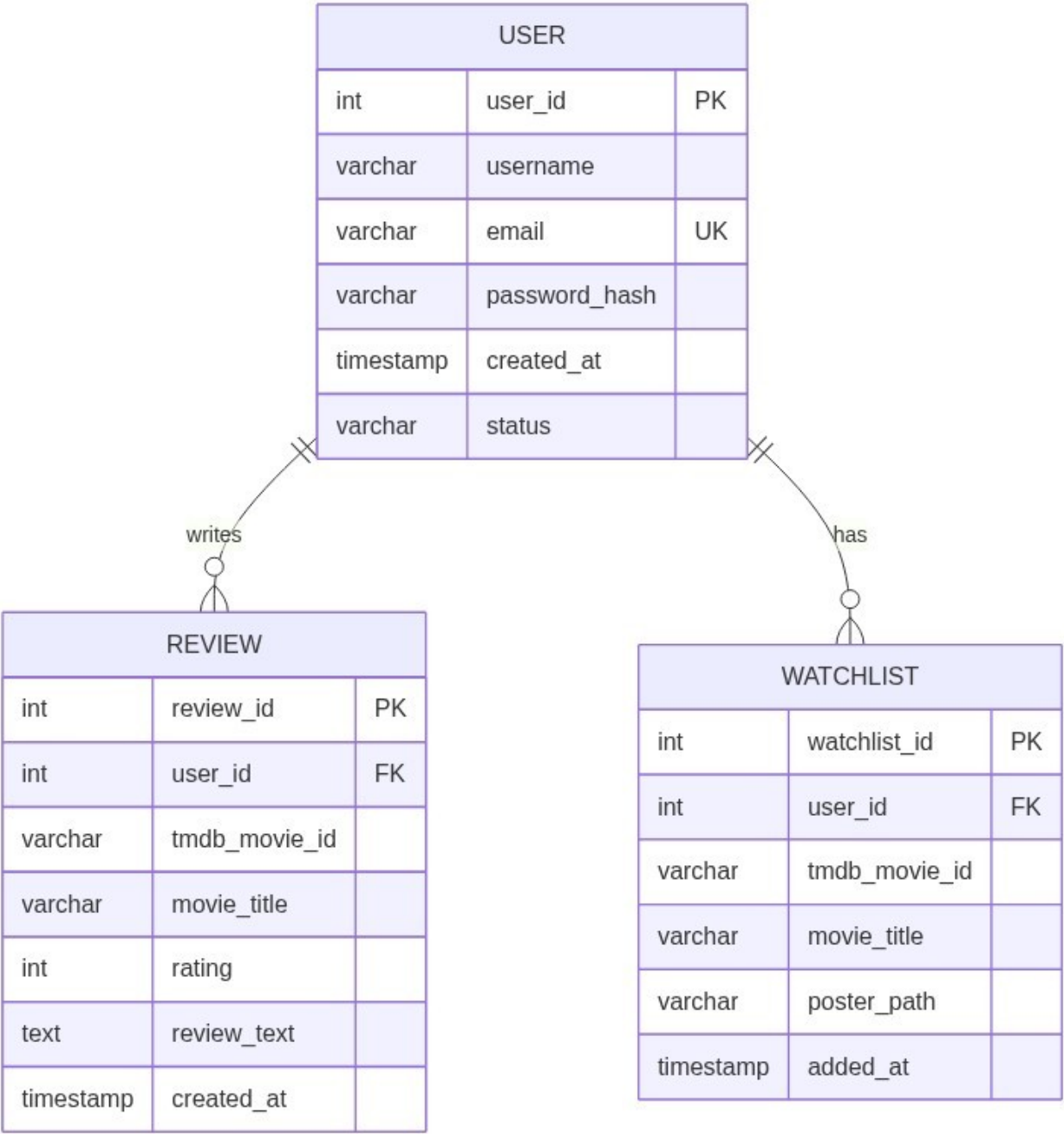
1.2 Relationships

Parent Entity	Card.	Child Entity	Card.	Description
USERS	1	WATCHLIST	M	A user can have many watchlist entries
USERS	1	REVIEWS	M	A user can write many reviews
USERS	1	WATCH_HISTORY	M	A user can have many watch history entries
GENRES	1	MOVIES	M	A genre can categorize many movies
MOVIES	1	WATCHLIST	M	A movie can be in many

				watchlists
MOVIES	1	REVIEWS	M	A movie can have many reviews
MOVIES	1	WATCH_HISTORY	M	A movie can be watched by many users

2. Document ER Diagram (MovieFlix Review-II Report Specification)

This ER diagram represents the database design as specified in the project review document (Section 12: ER Diagram). It uses TMDb movie IDs as external references rather than local movie storage.



2.1 Entity Definitions

USER

Column	Type	Key	Description
user_id	INT	PK	Auto-increment primary key
username	VARCHAR		Unique username
email	VARCHAR	UK	Unique email address
password_hash	VARCHAR		Securely hashed password
created_at	TIMESTAMP		Account creation timestamp
status	VARCHAR		Account status (active/inactive)

REVIEW

Column	Type	Key	Description
review_id	INT	PK	Auto-increment primary key
user_id	INT	FK	References USER.user_id
tmdb_movie_id	VARCHAR		TMDB external movie ID
movie_title	VARCHAR		Cached movie title for display
rating	INT		Rating 1-10
review_text	TEXT		Review content
created_at	TIMESTAMP		Review creation timestamp

WATCHLIST

Column	Type	Key	Description
watchlist_id	INT	PK	Auto-increment primary key
user_id	INT	FK	References USER.user_id
tmdb_movie_id	VARCHAR		TMDB external movie ID
movie_title	VARCHAR		Cached movie title
poster_path	VARCHAR		TMDB poster path
added_at	TIMESTAMP		When added to watchlist

2.2 Relationships

Parent Entity	Card.	Child Entity	Card.	Description
USER	1	REVIEW	M	A user can write many reviews
USER	1	WATCHLIST	M	A user can have many watchlist entries

3. Comparison: Implementation vs Document

Aspect	Document Specification	Actual Implementation
Movie Storage	External only (TMDB API)	Local MOVIES table + TMDB cache
Genres	Not specified	Dedicated GENRES table with FK
Watch History	Not specified	WATCH_HISTORY table with auto-update
Review PK	review_id (surrogate)	Composite (user_id, movie_id)
Watchlist PK	watchlist_id (surrogate)	id (surrogate)
Movie Reference	tmdb_movie_id (VARCHAR)	movie_id (VARCHAR) + local MOVIES.id
Cascading Deletes	Not specified	WATCHLIST.user_id ON DELETE CASCADE
Rating Precision	INT (1-10)	DECIMAL(2,1) for reviews, DECIMAL(3,1) for movies

3.1 Key Differences

- The implementation includes a local MOVIES table and GENRES table for richer local data storage and filtering capabilities.
- WATCH_HISTORY table tracks user viewing activity with automatic timestamp updates on re-watch.
- The document design relies entirely on TMDB API for movie metadata; the implementation caches key data locally.
- Composite primary keys in REVIEWS and WATCH_HISTORY prevent duplicate entries per user-movie pair.
- The implementation uses ON DELETE CASCADE on watchlist for referential integrity.